

Assignment

Course: B.Tech | Stream: Computer Science | Subject: Discrete Mathematics | Sem: x | Paper Code: yyy

Unit 1

Topics: Sets, Relations and Functions; Countability; Mathematical Induction; Number Theory | Total Marks: 20 | Total Time: 45 minutes

- Q1.** Define an equivalence relation and give one example. [2 Marks, L1]
- Q2.** Explain the difference between injective, surjective, and bijective functions, with examples. [3 Marks, L2]
- Q3.** Using the Euclidean Algorithm, find $\gcd(252, 105)$ and show all steps. [5 Marks, L3]
- Q4.** Analyze why the set of real numbers $\mathbb{R}$ is uncountable while the set of rational numbers $\mathbb{Q}$ is countable, referring to Cantor's diagonal argument. [5 Marks, L4]
- Q5.** A relation R is defined on the integers as aRb if $a \equiv b \pmod{5}$. Evaluate whether R is an equivalence relation, justifying your answer by verifying reflexivity, symmetry, and transitivity. [5 Marks, L5]